

# Testing Strategy

## Document:

docs/architecture/TESTING\_STRATEGY.md

**System:** HackNest — Student Operating System

**Status:** Authoritative — reconciled against

README.md , MASTER\_ARCHITECTURE.md ,

DATABASE\_ARCHITECTURE.md ,

SECURITY\_ARCHITECTURE.md , UI\_DESIGN\_SYSTEM.md ,

MASTER\_GOVERNANCE\_AMENDMENT\_001.md

**Type:** Architecture Specification Only — No Test Code

Per MASTER\_ARCHITECTURE.md Section 9, this document is the sole authoritative owner of testing levels, quality gates, and bug severity. Performance numeric targets are sourced directly from MASTER\_GOVERNANCE\_AMENDMENT\_001.md Section 4.5–4.6, closing the previously-flagged gap.

---

## 1. Purpose

Testing exists to make confidence in a release something HackNest can state with evidence. Every domain in DATABASE\_ARCHITECTURE.md and every

principle in `SECURITY_ARCHITECTURE.md` has a corresponding validation expectation, defined here.

---

## 2. Testing Principles

Shift Left, Test Pyramid, Automation First, Repeatability, Reliability, Performance Awareness, Security Awareness — unchanged from prior specification; still fully consistent with `MASTER_ARCHITECTURE.md` Section 2.

---

## 3. Testing Levels

Unit, Integration, API, Database, Authentication, Authorization, UI, Accessibility, Performance, Security, End-to-End, Regression, Smoke — each validating the corresponding subsystem document's contract ( `API_ARCHITECTURE.md` , `DATABASE_ARCHITECTURE.md` , `SECURITY_ARCHITECTURE.md` , `UI_DESIGN_SYSTEM.md` ) rather than redefining it.

---

## 4. Test Environment / Test Data Strategy

Local, Preview, Staging, Production – isolation and promotion per `DEPLOYMENT_ARCHITECTURE.md` . Test users/organizations/universities/hackathons/projects seeded per domain, with a defined reset cadence between runs – unchanged in substance from prior specification.

---

## 5. Performance Testing (Reconciled – Concrete Targets)

Sourced from `MASTER_GOVERNANCE_AMENDMENT_001.md` Section 4.5–4.6:

| Interaction Class                 | Target (p95)          |
|-----------------------------------|-----------------------|
| Interactive API reads             | < 300ms               |
| Search queries                    | < 500ms               |
| Mutating actions (Server Actions) | < 500ms               |
| Page navigation (server-rendered) | < 1.5s to interactive |
| File upload acknowledgment        | < 1s                  |

**Concurrency targets:** 50,000 concurrent sessions steady-state; sustain 5x (250,000) burst without exceeding 2x the above targets. **Scalability testing** validates behavior approaching the 1,000,000-student target and the partitioning triggers defined in `DATABASE_ARCHITECTURE.md` Section 13 (50 million rows or 2x target query time, whichever first).

---

## 6. Security Testing

Authentication validation, Authorization validation, Input validation, SQL injection prevention, XSS validation, CSRF validation, Rate limiting verification — each validated against the concrete rate-limit tiers now defined in `API_ARCHITECTURE.md` Section 6, rather than an undefined placeholder.

---

## 7. Accessibility Testing (Reconciled — Concrete Target)

**WCAG Goal: WCAG 2.2 Level AA**, cited directly from `UI_DESIGN_SYSTEM.md` Section 9 rather than independently deferred — closing the duplication `ARCHITECTURE_VALIDATION_REPORT.md` identified. Keyboard navigation, screen reader, color contrast,

and focus management testing all validate against that same target.

---

## 8. Release Quality Gates

Before Staging → Production promotion: all automated tests pass; no open Critical/High defects (Section 9); security testing complete with no unresolved findings; accessibility testing meets WCAG 2.2 AA; performance targets (Section 5) met or explicitly waived with documented justification; smoke testing passed on Staging immediately prior.

---

## 9. Bug Severity

Critical / High / Medium / Low, with remediation SLAs now concrete per

`MASTER_GOVERNANCE_AMENDMENT_001.md` Section 4.7:

Critical 24 hours, High 7 days, Medium 30 days (Low tracked in normal backlog).

---

## 10. CI/CD Testing & Reporting

Required automated checks (unit, integration, type-check, lint per `CODING_STANDARDS.md`), required

manual reviews for security-sensitive changes, build/deployment validation, and test/coverage/performance/security reporting — unchanged in substance, now grounded in concrete targets rather than placeholders throughout.

---

## Definition of Done

- [x] Architecture only — no test code.
  - [x] Reconciled against MASTER\_ARCHITECTURE.md , DATABASE\_ARCHITECTURE.md , SECURITY\_ARCHITECTURE.md , UI\_DESIGN\_SYSTEM.md , MASTER\_GOVERNANCE\_AMENDMENT\_001.md .
  - [x] Performance/concurrency targets and WCAG level now concrete, closing both previously-flagged gaps.
- 

## Validation Report

**Internal consistency:** confirmed — Section 5's targets and Section 7's WCAG citation are each stated once and referenced elsewhere without restatement.

**Cross-document consistency:**

performance/concurrency values match

MASTER\_GOVERNANCE\_AMENDMENT\_001.md Section 4.5–

4.6 verbatim; WCAG target now correctly cites  
UI\_DESIGN\_SYSTEM.md Section 9 as sole owner  
rather than independently deferring. **Resolved audit  
findings:** both the performance-target gap and the  
WCAG duplication flagged in  
ARCHITECTURE\_VALIDATION\_REPORT.md are closed.

**Status: Validated — Reconciled — Ready to Govern  
Implementation**